Deep Symmetric Fusion Transformer for Multimodal Remote Sensing Data Classification

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***Abstract*— In recent years, multimodal remote sensing data classification (MMRSC) has evoked growing attention due to its more comprehensive and accurate delineation of Earth's surface compared to its single-modal counterpart. However, it remains challenging to capture and integrate local and global features from single-modal data. Moreover, how to fully excavate and exploit the interactions between different modalities is still an intricate issue. To this end, we propose a novel dual-branch transformer-based framework named deep symmetric fusion transformer (DSymFuser). Within the framework, each branch contains a stack of local–global mixture (LGM) blocks, to extract hierarchical and discriminative single-modal features. In each LGM block, a local–global feature mixer with learnable weights is specifically devised to adaptively aggregate the local and global features extracted with a convolutional neural network (CNN)–transformer network. Furthermore, we innovatively design a symmetric fusion transformer (SFT) that trails behind each LGM block. The elaborately designed SFT symmetrically facilitates cross-modal correlation excavation, comprehensively exploiting the complementary cues underlying heterogeneous modalities. The hierarchical construction of the LGM and SFT blocks enables feature extraction and fusion in a multilevel manner, further promoting the completeness and descriptiveness of the learned features. We conducted extensive ablation studies and comparative experiments on three benchmark datasets, and the experimental results validated the effectiveness and superiority of the proposed method. The source code of the proposed method will be available publicly at https://github.com/HaixiaBi1982/DSymFuser.**

Received 24 March 2024; revised 14 September 2024; accepted 7 October 2024. Date of publication 14 October 2024; date of current version 21 October 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFA1003800, in part by NSFC under Grant 42201394 and Grant 12326615, and in part by the Qinchuangyuan Talent Program under Grant QCYRCXM-2022-30. *(Corresponding author: Haixia Bi.)*

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Digital Object Identifier 10.1109/TGRS.2024.3476975

***Index Terms*— Land-cover classification, local–global mixture (LGM), multimodal feature fusion, remote sensing, symmetric fusion transformer (SFT).**

I. Introduction

LAND-COVER classification, which aims at allocating pixel-wise terrain classes, is a fundamental task in the remote sensing community. It has been successfully applied and holds promising prospects in widespread applications, including atmospheric monitoring [1], urban planning [2], forest monitoring [3], [4], and environment monitoring [5]. With the rapid development of remote sensing technology, recent years have witnessed an unprecedented increase of remote sensing data, in both volume and diversity, which offers new research opportunities for land-cover classification [6], [7].

Over the past decades, massive efforts have been made in extracting powerful features and devising effective classifiers for the studied task. Nevertheless, these methods are mainly focusing on classifying data of single modalities, such as hyperspectral images (HSIs) [8], light detection and ranging (LiDAR) [9], and synthetic aperture radar (SAR) [10]. Despite the notable achievements, the descriptive capacity of single-modal data on the land cover remains limited. HSIs capture data over hundreds of narrow spectral bands, which is particularly advantageous in distinguishing different materials, even with subtle spectral differences. However, it is difficult for HSIs to discern two objects that are made of the same material, such as roads and roofs both made of cement, for their spectral correspondences are indiscernible in the HSI modality. Fortunately, with the height-sensitive LiDAR data incorporated, the above issue can be precisely solved [11], [12], [13]. In addition, optical imaging is susceptible to cloud obstruction during the acquisition process, resulting in partial information loss. Nevertheless, SAR is characterized by its all-time and all-weather traits, which makes it particularly useful in scenarios with unsatisfactory weather and light conditions [14]. Therefore, SAR can serve as a complementary modality of optical data to circumvent this dilemma. In conclusion, multiple unique but complementary remote sensing data can be integrated to improve the accuracy and reliability of land cover classification [15], [16].

In recent years, deep learning (DL) has exhibited great potential in multimodal remote sensing data classification

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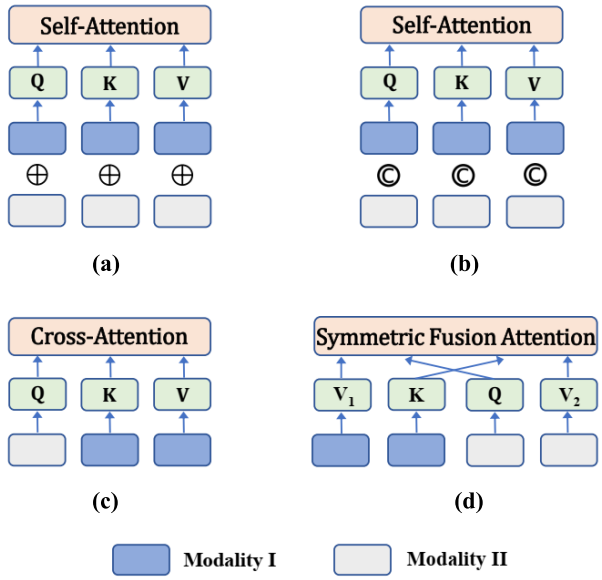


Fig. 1. Schematic illustration of four different transformer-based feature fusion strategies. (a) Self-attention (add). (b) Self-attention (concat). (c) Cross-attention. (d) SFA strategy.

(MMRSC) tasks. A corpus of researchers have investigated the application of convolutional neural networks (CNNs) in joint classification with multimodal data. Morchhale et al. [17], Ghamisi et al. [18], and Ge et al. [19] combined conventional handcrafted features with deep features learned via CNNs. Chen et al. [20], Deng et al. [21], Li et al. [22], and Audebert et al. [23] designed dual-branch or multiple-branch CNN architectures to independently learn representations from diverse modalities and then concatenate the learned features. The above CNN-based methods have remarkably boosted the classification performance owing to their strong capacity in local spatial feature learning, however, they have a limited ability to model global relationships.

Most recently, transformer networks [24] have been introduced to the field of MMRSC owing to its distinctive and potent long-range dependency modeling ability, either for single-modal feature extraction or multimodal heterogeneous feature fusion [25], [26], [27], [28], [29]. Considering how transformers are applied in drawing cross-modal interactions, the above methods can be grouped into three categories. Ding et al. [25] adopted a fusion strategy as shown in Fig. 1(a), where embeddings from two modalities are first extracted and added and then fed into a transformer layer for fusion via a self-attention mechanism. Zhang et al. [27], Yao et al. [28], and Tu et al. [30] aggregated multimodal data following the fusion architecture as illustrated in Fig. 1(b), where the input features for transformers are concatenated instead of being added. Roy et al. [31] and Zhao et al. [26] devised cross-attention-based feature fusion approaches, where the queries are taken as embeddings from one fixed modality, and keys and values come from the other, as exhibited in Fig. 1(c).

Investigating the CNN-based and transformer-based MMRSC approaches, we contend that developing an effective transformer-based cross-modal data classification method is two critical issues that need to be solved.

1) *How to acquire discriminative features?* From the above analysis, it can be inferred that the CNN is advantageous in capturing local spatial details via convolutions performed in small receptive fields, while the transformers excel in extracting global information by long-range attention. With only one network architecture involved, the learned representation would be incomplete in descriptiveness. Therefore, it is crucial to figure out an effective solution to combine the CNN and transformer, to obtain discriminative global–local features. Some MMRSC approaches [25], [26], [30] have delved into this problem. For example, Ding et . [25] designed aggregated CNNs to extract and fuse multimodal local spatial features from HSI-LiDAR data. Then, they used a transformer to learn global spectral features from HSI data and performed decision fusion based on separate predictions obtained with local and global features. Zhao et al. [26] sequentially extracted local and global features with CNNs and transformers. Tu et al. [30] first obtained local and global features from fused HSI-LiDAR features independently and then concatenated them for consequent classification. However, these works suffer from insufficient fusions between local and global features. We argue that local–global feature fusions, from both intramodality and intermodality aspects, are important for the classification. To this end, we design a local–global mixture (LGM) block, to capture intramodality local–global correlations. LGM blocks are further stacked, with information from other modalities injected, enabling multilevel intermodality local–global interaction excavation.
2) *How to fully excavate the interactions among multimodal data?* Existing cross-attention-based multimodal remote sensing data fusion strategy [26], [31] works in an unbalanced manner [see Fig. 1(c)], where one fixed modality acts as a query provider, and the other modality proffers keys and values. However, we argue that cross-attention in the opposite direction also contains inherent cross-modal relationships. Apart from direct utilization of the cross-attention block as shown in Fig. 1(c), some works [32], [33] combined multiple cross-attention blocks for feature fusion. However, they still put one modality in a disadvantaged position. This motivates us to develop a more comprehensive cross-modal feature fusion strategy, to fully leverage the interplay among modalities and further improve the performance of joint classification.

In response to the above motivations, we propose a novel dual-branch transformer-based framework for MMRSC, named deep symmetric fusion transformer (DSymFuser). Each modality branch contains a stack of LGM blocks, to extract hierarchical and discriminative single-modal features. In each LGM block, local and global features are first extracted with a CNN–transformer network, and a newly devised local–global feature mixer trails behind to adaptively aggregate them with learnable weights. Furthermore, to fully excavate the interactions between multimodal features, we innovatively design a cross-modal fusion strategy as illustrated in Fig. 1(d),

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which we call symmetric fusion attention (SFA). The uniqueness of SFA lies in that, unlike the original cross-attention which works in a fixed query fashion, SFA operates symmetrically, where each modality possesses an equal position, and alternately provides queries in the fusion process. SFA is followed by a feed-forward layer, forming a symmetric fusion transformer (SFT) block, which is performed after each LGM block, to excavate cross-modal relationships from different levels. To sum up, the main contributions of this article are as follows.

1) We propose a novel transformer-based classification framework, called DSymFuser, for the joint classification of multimodal remote sensing data. DSymFuser takes full advantage of the transformer network, achieving local–global feature extraction and cross-modal feature interaction in a multilevel manner, which is a significant endeavor to explore the potential of transformers in the studied task.
2) We innovatively design an SFT for cross-modal interaction excavation, which probes into the multimodal correlations in a symmetric query manner, maximizing the complementary advantages of each modality.
3) We design an LGM block, in which a transformer is incorporated with a lightweight CNN, to adaptively integrate intramodality local and global features. The multilevel design further enables hierarchical intermodality local and global feature fusion.
4) We conducted extensive experiments on three benchmark multimodal remote sensing classification datasets. The ablation study validated the effectiveness of each key component of the proposed method. Comparative experiments demonstrated that our proposed DSymFuser achieves state-of-the-art performance compared with eight competing methods.

The remainder of this article is organized as follows. The related works are recalled in Section II. We delineate the proposed method in Section III. Section IV reports the experimental data, results, and analysis. We finally conclude this article with future research prospects in Section V.

II. Related Works

A. Traditional Methods

Traditional MMRSC methods are primarily based on manually designed feature extractors. A great number of scholars have applied signal-processing operators and methods such as morphological profiles (MPs) [34], [35], attribute profiles (APs) [36], [37], extinction profiles (EPs) [38], wavelet filters [39], gray-level co-occurrence matrix (GLCM) [40], independent component analysis (ICA) [41], and sparse representation [42], for feature extraction on multimodal data. In addition, researchers have also designed feature extractors tailored for specific remote sensing data, such as spectral features for HSIs and scattering parameter features for SAR images [41].

As for the feature fusion, Liao et al. [34] proposed a graph-based fusion method to combine spectral information from HSI data with MPs from both HSI and LiDAR data. Hong et al. [43] explored the application of subspace-regularized learning in MMRSC. They proposed a shared and specific feature learning model, which decomposes multimodal remote sensing data into modality-shared and modality-specific components, facilitating the effective blending of information from diverse sources for improved land-cover classification. Pournemat et al. [44] simultaneously learned low-dimensional manifolds within each modality and aligned them across modalities, utilizing limited prior knowledge and localized Laplacian eigenmaps for semi-supervised learning. However, these conventional methods heavily rely on manually crafted features, failing to automatically capture the hidden information within the data.

B. Conventional DL Methods

With the burgeoning of DL techniques, representative deep architectures, especially CNNs, have been widely applied in multimodal land-cover classification. Gao et al. [45] proposed an ACL-CNN model, utilizing adversarial complementary learning to effectively extract complementary information from multisource remote sensing data. Hang et al. [46] designed a two-coupled CNN model to fuse HSI and LiDAR data, in which both feature-level and decision-level fusion methods are used to effectively integrate heterogeneous features. In [47], a novel dual-channel spatial–spectral and multiscale attention convolutional network is proposed for the classification of multisource remote sensing data. Wu et al. [48] put forward a cross-channel reconstructed convolutional neural network, leveraging cross-modal reconstruction strategies to enhance the classification performance.

Compared to traditional methods, DL-based approaches can automatically learn discriminative features, presenting promising results in multimodal land-cover classification tasks. Although the CNN has natural advantages in extracting local spatial information, it suffers from deficiencies in capturing the global relational characteristics of remote sensing data.

C. Transformer-Based Methods

Most recently, transformer networks [24] have garnered widespread attention from researchers in the field of remote sensing image processing, due to their powerful ability to model long-sequence relationships. To be specific, Sun et al. [49] developed a spectral–spatial feature tokenization transformer (SSFTT) model, which integrates spectral–spatial feature extraction with transformer-based semantic feature representation to improve HSI classification performance while reducing computational costs. Roy et al. [50] put forward morphFormer, which implements spectral and spatial morphological convolution operations combined with the attention mechanism to enhance interactions between structural and shape information in HSI classification.

Although the aforementioned methods have achieved outstanding performance, they are designed for single-modal data classification. Recently, researchers have successfully applied transformers to multimodal land-cover classification tasks. Wang et al. [51] introduced a multiattentive hierarchical

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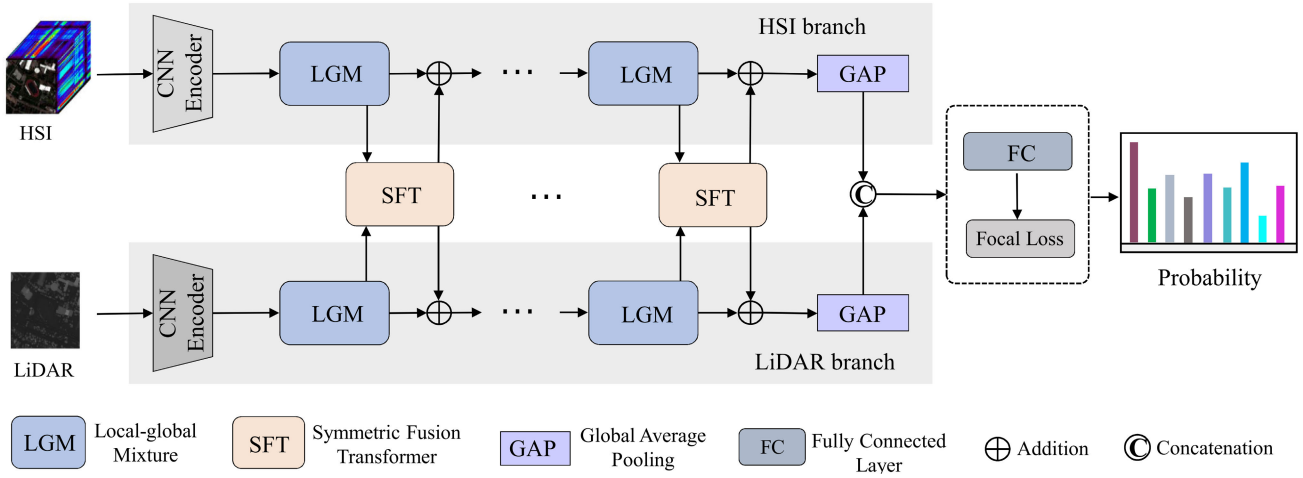


Fig. 2. Illustration of the proposed DSymFuser for MMRSC. DSymFuser is a dual-branch framework with LGT blocks stacked for hierarchical local and global feature learning and aggregation. SFT blocks are intersected between two branches for multilevel cross-modal feature fusion. The fused features are then fed into a classification block for the final probability prediction.

fusion network which includes a triple-branch CNN backbone and hierarchical fusion strategy with modality attention and self-attention modules for effective feature integration. A novel global–local transformer network (GLT-Net) was proposed in [25], which uses CNNs for local spatial feature learning and vision transformers (ViTs) for global spectral feature learning. Zhao et al. [26] proposed a novel dual-branch method for joint HSI-LiDAR data classification, with each branch combining a hierarchical CNN and a transformer, where the CNN and transformer are sequentially structured to extract local and global features. Inspired by the promising performance of ViT [31], [52] established a novel multimodal fusion transformer (MFT), which incorporates a multihead cross-patch attention mechanism for joint use of hyperspectral and other data sources. Zhao et al. [27] built a multimodal transformer network (MTNet), which integrates hyperspectral spectral transformer and HS-LiDAR spatial transformers to capture both spectral and spatial characteristics. Yao et al. [28] designed an extended ViT (ExViT), which utilizes position-shared ViT to process remote sensing image patches, followed by cross-modality attention fusion modules for multimodal heterogeneous feature fusion. Tu el al. [30] proposed to use structure information modeling transformer and invertible neural networks to fuse multisource remote sensing data.

Analyzing the above literature, we have the following findings: 1) for feature extraction, existing methods do not fully explore the interactions between local and global features and 2) for cross-modal interaction, current fusion approaches treat different modalities unfairly and neglect multilevel feature fusion. Our proposed DSymFuser differs from the above methods in below aspects: 1) our method takes into account the local–global interaction mining, which is implemented via a lightweight LGM block; 2) the novelly designed SFT fairly processes information from different modalities, sufficiently excavating the relationships between them; and 3) the feature extraction and cross-modal interaction operate in a multilevel fashion, enabling sufficient multimodal complementary information fusion.

III. Methodology

The task we investigate in this work is pixel-level classification using multimodal remote sensing data. Specifically, we consider the collaborative classification with HSI and LiDAR data in this article. Given a co-registered image pair, we define the input HSI data as $X_h \in \mathbb{R}^{h\times w\times c}$, and LiDAR data as $X_l \in \mathbb{R}^{h\times w\times 1}$, where h and w denote the height and width of the images, respectively, and c denotes the spectral channel numbers of HSIs. For each pixel, a pair of pixel-centered HSI image cube $x_h \in R^{p\times p\times c}$ and LiDAR image patch $x_l \in R^{p\times p\times 1}$ act as input, where p denotes the patch size. Our objective is to assign a class label y for each pixel, where $y \in \{1, 2, \ldots, K\}$, K is the total number of classes.

In the following, we will first describe the overall architecture of the proposed multimodal classification framework in Section III-A. Then, we will detail the key components of the proposed method in Sections III-B–III-E.

A. Overview

Fig. 2 illustrates our proposed DSymFuser, which is a dual-branch architecture with SFT blocks intersected for intermodality correlation mining. In each branch, shallow local features are first extracted using a CNN encoder, and then a string of LGM blocks is followed to extract multilevel local–global features. SFT blocks are performed after each LGM block pair, to mine cross-modal interactions between modalities, which act as input for the next layer of LGM blocks. This multilevel design enables multimodal correlation excavation from low-level features to high-level semantics, fully exploiting the complementary cues underlying different modalities. Finally, the features that have undergone multilevel feature learning and cross-modal information communication are concatenated and fed into a fully connected (FC) layer for classification.

Next, we would like to show how the data are processed in the pipeline. Given the input HSI cube x_h and LiDAR patch x_l, CNN encoders are first used to extract shallow local features

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Algorithm 1 Pseudocode of DSymFuser

Input: HSI image X_h, LiDAR data X_l, epochs E, depth of model L
Output: Classification map M

```
Create HSI cube x_h and LiDAR patch x_l from X_h and X_l;
Initialize all parameters;
while e < E do
    Extract shallow local features F_h^0 and F_l^0 following Eq. (1);
    while i < L do
        Extract ith layer local–global features F_h^i and F_l^i following Eq. (2);
        Generate ith layer cross-modal interaction features C^i following Eq. (3);
        Fuse cross-modal interaction features and single-modal features according to Eq. (4);
    end while
    Obtain the final output features F_end with Eq. (18);
    Get the predicted probability y_p following Eq. (19);
end while
Obtain the classification map M;
```

for two modalities as follows:

$$\mathcal{F}_h^0 = \mathcal{M}_h(x_h), \quad \mathcal{F}_l^0 = \mathcal{M}_l(x_l) \tag{1}$$

where $\mathcal{M}_h(\cdot)$ and $\mathcal{M}_l(\cdot)$ represent the CNN encoders for HSI and LiDAR patches, respectively, and $\mathcal{F}_h^0$ and $\mathcal{F}_l^0$ denote the shallow local features of HSI and LiDAR, respectively.

Afterward, LGM [denoted by $\mathcal{L}(\cdot)$] is performed to further extract local–global features for each modality

$$\mathcal{F}_h^i = \mathcal{L}\left(\mathcal{F}_h^{i-1}\right), \quad \mathcal{F}_l^i = \mathcal{L}\left(\mathcal{F}_l^{i-1}\right) \tag{2}$$

where i represents the ith layer. It should be noted that for the first LGM layer, the model takes $\mathcal{F}_h^0$ and $\mathcal{F}_l^0$ as input.

After each LGM layer, SFT [denoted by $\mathcal{S}(\cdot)$] is conducted to maximize the complementary effect across heterogeneous modalities. Specifically, the cross-modal information of the ith layer $\mathcal{C}^i$ is generated as

$$\mathcal{C}^i = \mathcal{S}\left(\mathcal{F}_h^i, \mathcal{F}_l^i\right). \tag{3}$$

$\mathcal{C}^i$ is further added to both the HSI branch and LiDAR branch, to boost the feature fusion of the next layer

$$\mathcal{F}_h^i = \mathcal{C}^i + \mathcal{F}_h^i, \quad \mathcal{F}_l^i = \mathcal{C}^i + \mathcal{F}_l^i. \tag{4}$$

As a combinational block, the process denoted by (2)–(4) is implemented for L times for hierarchical feature extraction and fusion, where L indicates the number of layers. The Lth layer features $\mathcal{F}_h^L$ and $\mathcal{F}_l^L$ are then fed to global average pooling (GAP) layers, where the outputs are concatenated as $\mathcal{F}_{\text{end}}$. $\mathcal{F}_{\text{end}}$ further passed through an FC layer for the final classification. For a clearer understanding of the workflow, we exhibit the pseudocode of the proposed DSymFuser in Algorithm 1. The key components of DSymFuser, including the CNN encoder, LGM, SFT, and classification blocks will be detailed consequently.

TABLE I
DETAILS OF THE CNN ENCODERS

Modality	Layer type	Kernel size	Output size
HSI	Input	—	(1, c, 13, 13)
	Conv3D	8@5×3×3	(8, c,11, 11)
	Reshape	—	(8×c, 11, 11)
	Conv2D	64@3×3	(64, 9, 9)
LiDAR	Input	—	(1, 13, 13)
	Conv2D	64@3×3	(64, 11, 11)
	Conv2D	64@3×3	(64, 9, 9)

B. CNN Encoder

The initial application of a transformer in the field of computer vision is scene-level natural image classification. In the extensively applied ViT [52], images are first divided into fixed-size patches, and then the linear embeddings are sequentially input into the transformer. However, when it comes to pixel-level classification tasks, this flattening operation does not apply anymore, for it will undoubtedly result in severe loss of contextual information. Nevertheless, the CNN is renowned for its superior ability to model local information. To this end, instead of directly feeding the input patches to the transformer, CNN encoders are employed in prior to extracting local features, to fully preserve and utilize local information contained in remote sensing data, such as textures and edges.

Specifically, for the HSI branch, considering the rich spectral information contained in HSI data, we first employ 3-D convolutions (Conv3D) to capture the spectral sequence characteristics of the HSI data. A 2-D convolution (Conv2D) layer is then followed to extract discriminative spatial features. For the LiDAR branch, two Conv2D layers are deployed to obtain local spatial features. The summary of the two branches in terms of the layer type, kernel size, and output size is shown in Table I, taking the Houston2013 dataset as an example. In addition, to enhance model stability and introduce nonlinearity, batch normalization (BN) and Gaussian error linear unit (GeLU) are consecutively applied after each convolution layer. It is worth noting that the utilized CNN encoders are generalized designs that can be seamlessly applied to the joint classification of HSIs and another modality whose input format is a 2-D image, such as SAR and infrared. Even for modalities that Conv2D and Conv3D do not apply, the whole pipeline will still work only by changing the CNN encoders to suit the data format of the new modality.

C. LGM Block

In MMRSC tasks, both CNN- and transformer-based models have been extensively investigated [18], [23], [30], [31]. These two categories of methods have their respective advantages and shortcomings. The CNN excels in capturing local information, while the transformer presents powerful capabilities in capturing global contextual information. Given the fact that both local and global information is crucial for remote sensing image representation, we plan to integrate the CNN and transformer to take advantage of their strengths,

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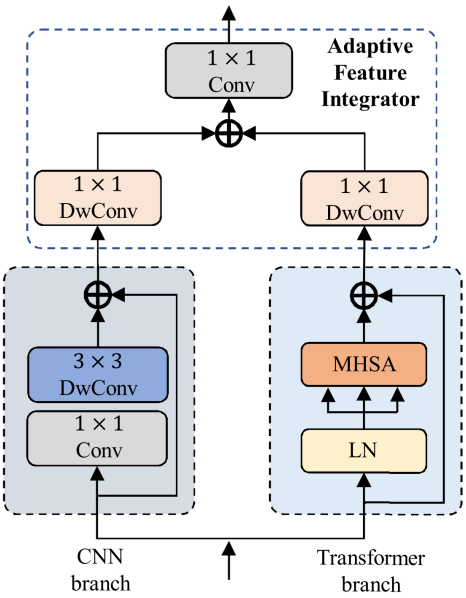


Fig. 3. Schematic illustration of the LGM block.

where there are two issues to be dealt with during the integration. The first is how to effectively fuse features from the CNN and transformer, and the second is how to reduce the model's complexity.

In this context, we design an LGM block, which takes a dual-to-single architecture, as shown in Fig. 3. The framework consists of three components, which are a *CNN branch* for local information extraction, a *Transformer branch* for global contextual information extraction, and an *adaptive feature integrator* for local–global feature fusion. The details of the three components will be depicted in the following paragraphs.

In the CNN branch, instead of the conventionally used convolutions, the CNN branch is built on lightweight separable convolutions [53]. Assuming C_{in} and C_{out} represent the input and output channel numbers of the feature map, respectively, and k represent the convolution kernel size, the number of parameters of the separable convolution is $C_{\text{in}} \times C_{\text{out}} + k \times k \times C_{\text{out}}$, while the number of parameters of the standard convolution is $k \times k \times C_{\text{in}} \times C_{\text{out}}$. Then, it can be inferred that the number of parameters of the separable convolution is much smaller than that of the standard convolution, especially when either k or C_{in} is large. Therefore, by replacing standard convolution with separable convolution, the number of parameters of the module is greatly reduced. Specifically, one separable convolution block is employed for local feature extraction, whose process is outlined as

$$\mathcal{F}_{\text{local}} = \text{DwConv}_{3\times3}(\text{Conv}_{1\times1}(\mathcal{F})) + \mathcal{F} \tag{5}$$

where $\mathcal{F}_{\text{local}}$ denotes the generated local feature, $\text{Conv}_{1\times1}$ is the pointwise convolution, and $\text{DwConv}_{3\times3}$ indicates the depthwise convolution with kernel size 3×3.

In the transformer branch, given the input feature $\mathcal{F}$, multihead self-attention (MHSA) is leveraged to model long-distance relationships and capture global features $\mathcal{F}_{\text{global}}$, which can be expressed as

$$\mathcal{F}_{\text{global}} = \text{MHSA}(\text{LN}(\hat{\mathcal{F}})) + \hat{\mathcal{F}} \tag{6}$$

where $\hat{\mathcal{F}}$ is the flattened feature of $\mathcal{F}$, MHSA$(\cdot)$ and LN$(\cdot)$ denote multihead self-attention and layer normalization, respectively. For each attention head, three learnable linear matrices W_q, W_k, and W_v project $\hat{\mathcal{F}}$ into Q_i, K_i, and V_i, respectively (i denotes the ith head). The ith-head self-attention SA_i can be calculated by the following equation:

$$\text{SA}_i(Q, K, V) = \text{Softmax}\left(\frac{Q_i K_i^{\text{T}}}{\sqrt{d/n}}\right) V_i \tag{7}$$

where T represents transpose operation, d denotes the channel dimension of V, and n is the number of attention heads. Finally, all attention heads are concatenated as the output.

The adaptive feature integrator aims to effectively fuse $\mathcal{F}_{\text{local}}$ and $\mathcal{F}_{\text{global}}$. Instead of simply adding or concatenating the output features from CNN and transformer branches, our designed feature integrator uses learnable parameters to adaptively coordinate the weights over local and global feature components, which can be expressed as

$$\mathcal{F}' = \text{DwConv}_{1\times1}(\mathcal{F}_{\text{global}}) + \text{DwConv}_{1\times1}(\mathcal{F}_{\text{local}}) \tag{8}$$

$$\mathcal{F} = \text{Conv}_{1\times1}(\mathcal{F}') \tag{9}$$

where $\text{DwConv}_{1\times1}$ is a depthwise convolution with kernel size 1×1 and $\mathcal{F}'$ is an intermediate variable. Specifically, each convolution layer is followed by a BN layer and the GeLU function. This approach essentially performs a channel-wise weighted summation of the features from two branches, achieving an adaptive aggregation of them.

D. SFT Block

Cross-modal feature interaction is pivotal for MMRSC, for integrating complementary information from different modalities can significantly improve the classification accuracy [54]. To fulfill the purpose of multimodal data fusion, a number of transformer-based fusion approaches [31] have been proposed. Since the attention mechanism is adept at mining intermodality relations, transformer-based methods demonstrate superior performance compared to traditional fusion methods such as direct concatenation or summation. However, existing transformer-based fusion methods mainly utilize cross-attention mechanisms to fuse information from different modalities, placing one modality in a disadvantaged position, which may result in multimodal correlation information missing. We argue that both modalities should be treated fairly, allowing the network to adaptively learn the optimal fusion strategy for multimodal features.

In this article, an SFT block is proposed, whose core concept is a novel SFA mechanism as shown in Fig. 4. Unlike previous transformer-based feature fusion methods [31], [32], [33], we do not simply adopt the standard single or multiple cross-attention blocks for feature fusion, but innovatively design an SFA, where different modalities are treated fairly, and the attention scores are calculated only once. SFT differs from the standard attention in two aspects. First, both modalities have equal chances to act as queries and values, yielding two respective values (V_1 and V_2 in Fig. 4) containing complementary information from the other modality. Second, to introduce local receptive field information while controlling

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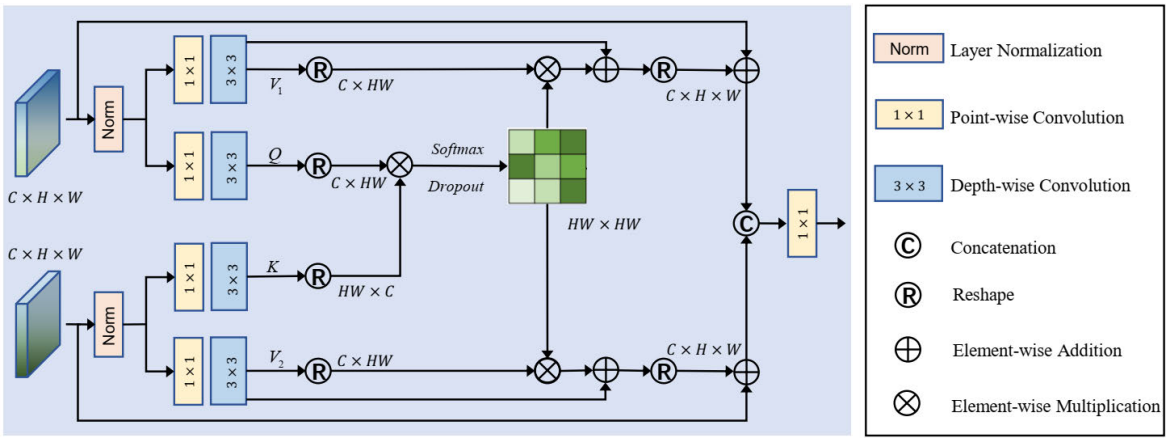


Fig. 4. Schematic illustration of the SFA.

the number of parameters, we utilize lightweight convolutions instead of linear layers to generate Q, K, and V. The process is shown as

$$Q = \text{Conv}_{1\times1}(\text{DwConv}_{3\times3}(\text{LN}(\mathcal{F}_h))) \tag{10}$$
$$K = \text{Conv}_{1\times1}(\text{DwConv}_{3\times3}(\text{LN}(\mathcal{F}_l))) \tag{11}$$
$$V_1 = \text{Conv}_{1\times1}(\text{DwConv}_{3\times3}(\text{LN}(\mathcal{F}_h))) \tag{12}$$
$$V_2 = \text{Conv}_{1\times1}(\text{DwConv}_{3\times3}(\text{LN}(\mathcal{F}_l))) \tag{13}$$

where $\mathcal{F}_h$ and $\mathcal{F}_l$ represent the input features of the HSI branch and LiDAR branch, respectively.

Given the query Q and key K, the cross-modal interaction map A is generated following:

$$A = \text{Dropout}\left(\text{Softmax}\left(\frac{Q^T K}{\alpha}\right)\right) \tag{14}$$

where α is a learnable scaling parameter that is used to control the magnitude of the dot product between Q and K. Dropout operation is applied to enhance the model's generalization ability, where the dropout rate is set to 0.1.

With the cross-modal interaction map A, we next infer HSI feature Ψ_h and LiDAR feature Ψ_l following:

$$\Psi_h = V_1 A + V_1 + \mathcal{F}_h \tag{15}$$
$$\Psi_l = V_2 A + V_2 + \mathcal{F}_l. \tag{16}$$

It is noteworthy that for each modality, a residual connection with the original feature input is incorporated as well to improve the stability of the model.

Finally, we obtain the cross-modal information $\mathcal{C}'$ by first concatenating Ψ_h and Ψ_l and then performing point-wise convolution on the concatenated features, where the process can be represented as

$$\mathcal{C}' = \text{Conv}_{1\times1}([\Psi_h; \Psi_l]) \tag{17}$$

where $[\cdot]$ denotes the concatenation operation.

E. Classification Block

After L rounds of local–global feature extraction and cross-modal feature fusion, we obtain the multimodal fused feature $\mathcal{F}_h^L$ and $\mathcal{F}_l^L$. Then, the final fusion feature $\mathcal{F}_{\text{end}}$ is obtained following:

$$\mathcal{F}_{\text{end}} = \left[\text{GAP}\left(\mathcal{F}_h^L\right); \text{GAP}\left(\mathcal{F}_l^L\right)\right] \tag{18}$$

where GAP represents the global average pooling operation. Followed by an FC layer and a Softmax layer, the fusion feature $\mathcal{F}_{\text{end}}$ is converted into the output probability y_p, where the process can be written as

$$y_p = \text{Softmax}(\text{Linear}(\mathcal{F}_{\text{end}})) \tag{19}$$

where Linear$(\cdot)$ indicates the FC layer.

The MMRSC task has been afflicted with a positive–negative sample imbalance issue. Specifically, when the number of negative samples far exceeds that of the positive samples, the model would be biased toward easily classified negative samples, which is detrimental to the optimization of the loss function. To solve this problem, instead of the commonly applied cross-entropy loss function, we adopt focal loss [55] in this article, the formula of which is as follows:

$$\mathcal{L} = -\beta(1 - y_p)^\gamma \log(y_p) \tag{20}$$

where β is employed to balance the positive and negative sample losses and is set to 0.25. γ is a hyperparameter that is used to reduce the contribution of easily classified negative samples. It takes the value of 2.

IV. Experimental Results and Analysis

In this section, we evaluate the effectiveness of the proposed model on three widely used multimodal land-cover classification datasets, that is, Houston2013 [35], Trento [41], and MUUFL [56] datasets, whose key information and training–test sample distributions are displayed in Tables II–V, respectively. In this section, we first introduce the evaluation metrics and implementation details. Afterward, extensive comparative experiments with state-of-the-art methods are performed and analyzed, to validate the superiority of our method. Thereafter, we validate the effectiveness of the key components of our model through ablation studies. Finally, we analyze the effects of different parameters on the classification performance.

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TABLE II
KEY INFORMATION OF THE EXPERIMENTAL DATASETS

Dataset	Houston	Trento	MUUFL
Location	Houston	Trento	Long Beach
Sensor type	HSI,DSM	HSI,DSM	HSI,DSM,DEM
Channel	144,1	63,1	64,2
Image size	349×1905	600×166	325×220
Train samples	2832	819	1650
Test samples	12197	29595	52037
Classes	15	6	11

TABLE III
TRAINING–TEST SAMPLE DISTRIBUTION FOR THE HOUSTON2013 DATASET

No.	Class Name	Train	Test
1	Healthy grass	198	1053
2	Stressed grass	190	1064
3	Synthetic grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot 1	192	1041
13	Parking lot 2	184	285
14	Tennis court	181	247
15	Running track	187	473
	Total	2832	12197

TABLE IV
TRAINING–TEST SAMPLE DISTRIBUTION FOR THE TRENTO DATASET

No.	Class Name	Train	Test
1	Apple trees	129	3905
2	Buildings	125	2778
3	Ground	105	374
4	Wood	154	8969
5	Vineyard	184	10317
6	Roads	122	3052
	Total	819	29595

TABLE V
TRAINING–TEST SAMPLE DISTRIBUTION FOR THE MUUFL DATASET

No.	Class Name	Train	Test
1	Trees	150	23096
2	Mostly grass	150	4120
3	Mixed ground surface	150	6732
4	Dirt and sand	150	1676
5	Road	150	6537
6	Water	150	316
7	Building shadow	150	2083
8	Building	150	6090
9	Sidewalk	150	1235
10	Yellow curb	150	33
11	Cloth panels	150	119
	Total	1650	52037

A. Experimental Setup

1) Evaluation Metrics: To evaluate the classification performance of the proposed method and other state-of-the-art competitors, three extensively used quantitative metrics are selected, which are overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa), respectively. Specifically, OA represents the proportion of correctly classified test samples to all test samples, while AA denotes the AA over all classes. Kappa reflects the consistency between the classification map generated by the applied model and the ground-truth map. For each metric, a higher value indicates better classification performance.

2) Implementation Details: To optimize the network, the Adamw optimizer is selected, with a weight decay parameter set to 0.01. During training, the initial learning rate is set to 0.0008, and a cosine annealing strategy is adopted to decay the learning rate per epoch. In addition, we train the model for a total of 100 epochs for the Trento dataset and 150 epochs for the Houston2013 and MUUFL datasets with a batch size of 16. All the experiments were implemented in the PyTorch framework on an OmniSky SCS4450-G2 workstation with 4 GeForce RTX A5000 GPUs and 64-GB memory.

B. Classification Results and Analysis

To demonstrate the effectiveness of the proposed DSymFuser, we compared it with eight state-of-the-art methods, including EndNet [57], MDL-RS [54], CCR [48], CALC [58], GLT [25], HCT [26], MFT [31], and ExViT [28]. For a fair comparison, we determined all parameters of the compared methods according to the descriptions in the original papers or based on the default settings of the open-source codes.

1) Houston2013 Dataset: Table VI illustrates the quantitative results of different methods on the Houston2013 dataset, while the visualized classification maps are shown in Fig. 5. It can be discovered from Table VI that our proposed DSymFuser achieves the highest values in terms of all three criteria compared with other methods. Specifically, DSymFuser outperforms the second-best method, that is, HCT, by 1.54%, 1.02%, and 1.67%, on OA, AA, and Kappa, respectively. Moreover, we achieved the best classification accuracy on seven classes, which are highlighted in bold in the table. Among them, for *Commercial* and *Highway* classes, our method achieves classification accuracy of 96.58% and 96.04%, respectively, which are remarkably higher than other competitors. It is noticed that our proposed DSymFuser obviously lags behind HCT in classes *Healthy grass* and *Water*. Two reasons may account for this phenomenon. First, the spectral characteristics of *Healthy grass* and *Stressed grass* are similar, and the height difference between them is marginal. Therefore, the complementary elevation information provided by LiDAR data is not discriminative enough, leading to limited fusion efficacy. Second, *Healthy grass* and *Stressed grass* are usually adjacent, same as *Water* and *Road*. If the local receptive field of a network is large, perturbing information from adjacent pixels would be involved, increasing the

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TABLE VI
CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET. BEST RESULTS ARE SHOWN IN BOLD

Class	EndNet	MDL-RS	CCR	CALC	GLT	HCT	MFT	ExViT	DSymFuser
Healthy grass	85.00	80.01	83.09	81.96	83.10	**86.61**	83.09	82.15	81.86
Stressed grass	97.84	**100.0**	**100.0**	98.97	99.53	**100.0**	99.62	98.87	**100.0**
Synthetic grass	99.21	99.80	**100.0**	94.26	96.83	99.80	99.41	96.24	98.81
Tree	96.88	96.21	99.34	96.02	**99.81**	98.10	97.06	96.31	99.15
Soil	93.66	99.81	**100.0**	99.81	**100.0**	**100.0**	**100.0**	99.91	**100.0**
Water	95.80	**100.0**	95.80	98.60	97.20	**100.0**	95.80	98.60	95.80
Residential	86.75	78.54	91.42	90.11	93.94	92.16	88.15	79.57	**94.87**
Commercial	87.84	86.61	76.26	90.41	91.55	93.35	90.88	92.69	**96.58**
Road	69.69	88.67	87.35	95.18	**95.85**	94.43	93.48	84.23	95.00
Highway	66.22	94.31	94.21	92.57	82.14	88.51	93.63	78.57	**96.04**
Railway	96.96	93.74	93.07	99.53	97.34	**100.0**	99.53	96.68	98.10
Parking lot 1	89.43	84.92	85.98	90.87	94.04	87.03	89.91	78.19	**96.06**
Parking lot 2	73.68	**92.98**	88.42	85.96	84.91	91.93	90.18	91.23	91.93
Tennis court	**100.0**	98.79	**100.0**	98.38	97.17	**100.0**	**100.0**	98.79	98.79
Running track	98.73	**100.0**	99.79	99.15	**100.0**	95.77	**100.0**	99.79	**100.0**
OA	88.07	91.40	91.97	93.78	94.03	94.49	94.11	89.86	**96.03**
AA	89.18	92.96	92.98	94.12	94.23	95.18	94.72	91.45	**96.20**
Kappa	87.05	90.67	91.28	93.24	93.52	94.02	93.61	88.99	**95.69**

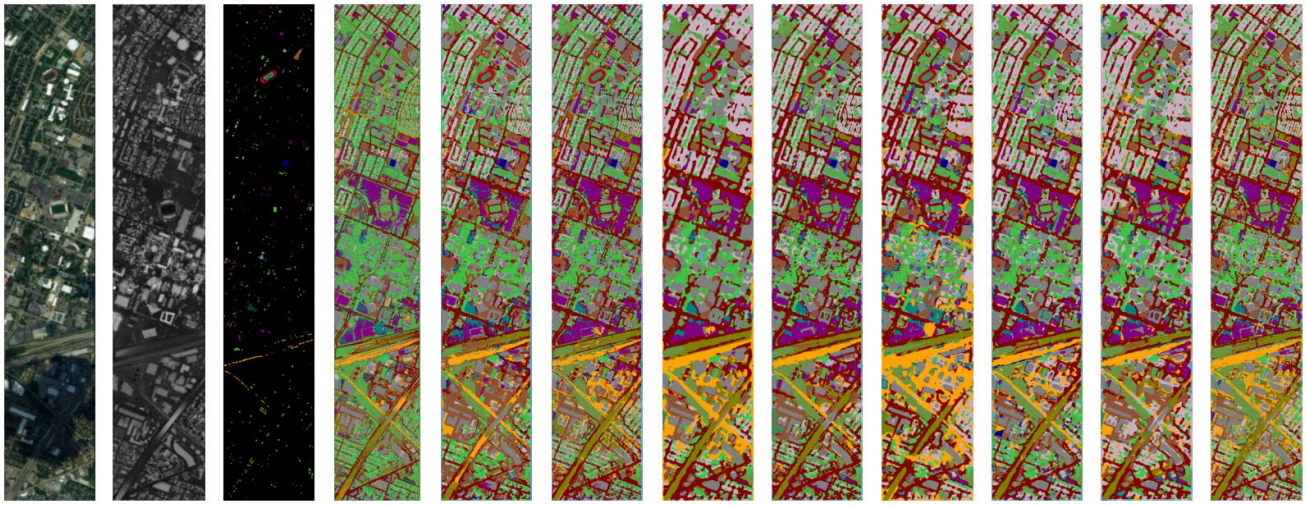


Fig. 5. Classification maps generated by different methods on the Houston2013 dataset. (a) Pseudo-color image for HSIs. (b) LiDAR-based DSM. (c) Ground-truth map. (d) ENdNet (88.07%). (e) MDL-RS (91.40%). (f) CCR (91.97%). (g) CALC (93.78%). (h) GLT (94.03%). (i) HCT (94.49%). (j) MFT (94.11%). (k) ExViT (89.86%). (l) Our proposed DSymFuser (96.03%).

risk of misclassification. The network depth of our proposed DSymFuser is larger than HCT, which signifies a larger local receptive field. This precisely explains the inferior classification performance of DSymFuser on *Healthy grass* and *Water*.

Observing Fig. 5(a), we can find that the bottom part of the area is obscured by cloud cover in the HSI image, making it challenging to discern the ground objects within it, especially the long-shaped *Highway*. Our proposed DSymFuser perfectly overcomes this hurdle, which can be substantiated by the clear boundaries of the *Highway*. The salient edges in the top half of the image are another evidence of the favorable performance of DSymFuser. This strongly indicates that our algorithm can effectively integrate complementary information from different modalities. Moreover, DSymFuser produces more fine-grained classification maps owing to the incorporation of local features.

2) Trento Dataset: The numerical results and the classification maps of different methods on the Trento dataset are shown in Table VII and Fig. 6, respectively. It can be found that our proposed model achieves the best classification performance with regard to OA, AA, and Kappa metrics. We notice that EndNet [57] does not behave well on the Trento dataset. This is because this method takes single pixels as the network input, which neglects contextual information during feature extraction. This proves that local neighboring information of pixels is crucial to the land-cover classification

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TABLE VII
CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE TRENTO DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class	EndNet	MDL-RS	CCR	CALC	GLT	HCT	MFT	ExViT	DSymFuser
Apple trees	75.62	98.98	**99.97**	98.03	99.10	98.92	98.59	97.59	99.54
Buildings	96.67	98.88	99.64	99.93	98.31	99.96	**100.0**	97.58	96.76
Ground	94.92	84.49	88.24	89.04	90.64	**97.06**	95.45	90.91	96.26
Wood	99.72	99.80	99.96	**100.0**	**100.0**	**100.0**	99.97	99.70	99.98
Vineyard	91.32	99.93	99.59	99.76	99.97	**100.0**	99.97	99.73	99.68
Roads	90.07	94.17	88.89	89.65	95.61	86.44	76.67	95.81	**97.80**
OA	92.22	98.87	98.50	98.43	99.14	98.41	97.31	98.71	**99.24**
AA	91.39	96.04	96.05	96.07	97.27	97.06	95.11	96.89	**98.34**
Kappa	89.56	98.49	97.99	97.90	98.84	97.87	96.40	98.28	**98.98**

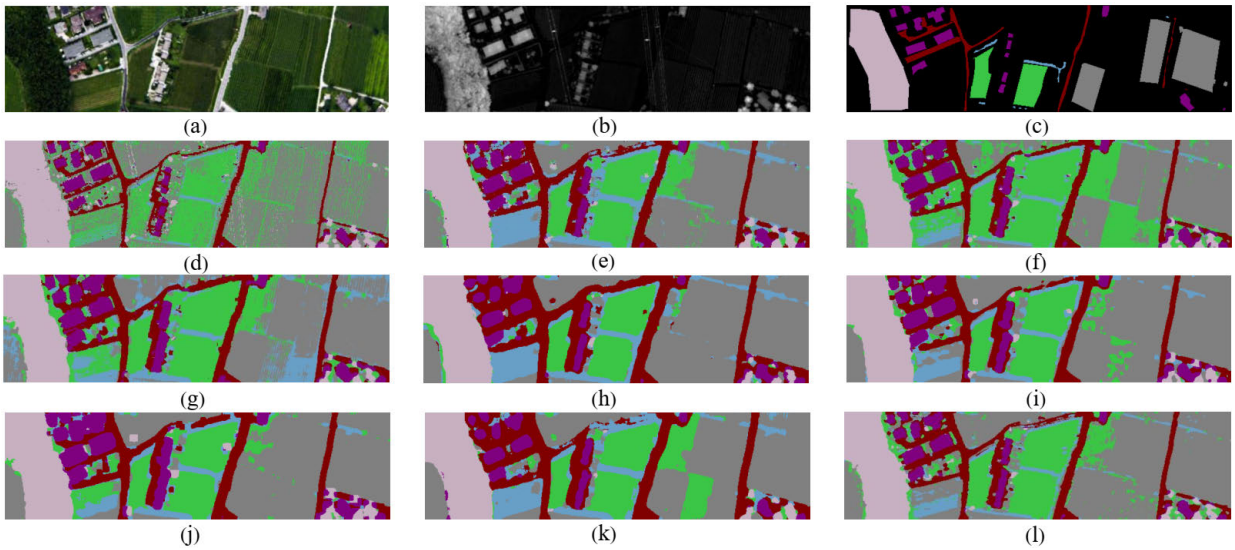


Fig. 6. Classification maps generated by different methods on the Trento dataset. (a) Pseudo-color image for HSIs. (b) LiDAR-based DSM. (c) Ground-truth map. (d) ENdNet (92.22%). (e) MDL-RS (98.87%). (f) CCR (98.50%). (g) CALC (98.43%). (h) GLT (99.14%). (i) HCT (98.41%). (j) MFT (97.31%). (k) ExViT (98.71%). (l) Our proposed DSymFuser (99.24%).

task, which is precisely coincident with our motivation to combine local–global features. The superiority of DSymFuser over other transformer-based methods confirmed the effectiveness of our proposed symmetric fusion scheme in integrating complementary multimodal information. Moreover, the proposed DSymFuser generates the best accuracy (97.90%) on *Road* category. It can also be seen from the figure that the edges of the *Road* are more compatible with the ground-truth map, compared with other transformer-based methods, which further verifies the advantage of combining local and global features.

3) MUUFL Dataset: Table VIII tabulates the numerical results of nine approaches on the MUUFL dataset. From the table, we can see that our proposed DSymFuser again yields the highest OA (93.32%), AA (94.01%), and Kappa (91.17%). Although DSymFuser does not achieve the highest classification accuracy in part of the categories, the accuracy differences between them and the highest values are marginal. Whereas, for challenging *Mixed ground surface* and *Buildings* classes, we obtain an accuracy of 90.46% and 96.60%, respectively, which presents a nontrivial advantage over the second-best result. The visual classification maps are displayed in Fig. 7. From the table and figure, we discovered that two transformer-based methods, that is, MFT [31] and ExViT [28], perform poorly on this dataset, both on the numerical results and visual classification maps. The reason may stem from the fact that these two methods pay limited attention to local information. Moreover, their intermodal interactions are either based on traditional self-attention or cross-attention mechanisms, where the cross-modal complementary information is not fully exploited. Whereas, our proposed DSymFuser demonstrates both a high resemblance with the ground-truth map and desirable preservation of details, such as textures and edges. This is attributed to two reasons. First, the local–global feature mixture takes both the local and global features into consideration. Second, our designed deep symmetric fusion strategy excavates more comprehensive cross-modal interactions, due to the symmetric query manner and hierarchical architecture.

C. Ablation Study

1) Ablation Analysis of Different Modal Inputs: To assess the influence of different modal inputs on the classification

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TABLE VIII
CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE MUUFL DATASET. BEST RESULTS ARE SHOWN IN BOLD

Class	EndNet	MDL-RS	CCR	CALC	GLT	HCT	MFT	ExViT	DSymFuser
Trees	**95.17**	91.63	92.63	93.98	93.35	94.60	89.40	89.71	94.55
Mostly grass	73.71	85.73	85.70	79.17	**88.76**	84.51	78.08	80.07	88.33
Mixed ground surface	76.02	83.27	75.83	84.15	85.37	86.47	75.36	67.86	**90.46**
Dirt and sand	83.95	92.00	90.51	91.17	**98.03**	93.74	96.84	92.30	97.97
Road	88.66	87.06	90.43	88.86	**92.50**	89.29	82.10	78.71	91.23
Water	99.37	99.37	99.68	**100.0**	**100.0**	**100.0**	**100.0**	99.68	99.68
Building shadow	70.62	86.94	95.49	**97.22**	95.54	92.03	90.45	89.58	94.34
Building	89.61	96.44	96.16	96.22	95.24	93.78	93.71	93.94	**96.60**
Sidewalk	81.70	79.27	**87.61**	79.03	86.23	80.32	63.32	51.42	86.96
Yellow curb	66.67	93.94	**96.97**	93.94	93.94	**96.97**	90.91	75.76	93.94
Cloth panels	97.48	99.16	99.16	99.16	**100.0**	**100.0**	99.16	97.48	**100.0**
OA	87.88	89.67	90.03	90.89	92.20	91.56	86.03	84.47	**93.32**
AA	83.91	90.44	91.84	91.17	93.54	91.97	87.21	83.32	**94.01**
Kappa	83.80	86.43	86.88	87.96	89.71	88.84	81.83	79.78	**91.17**

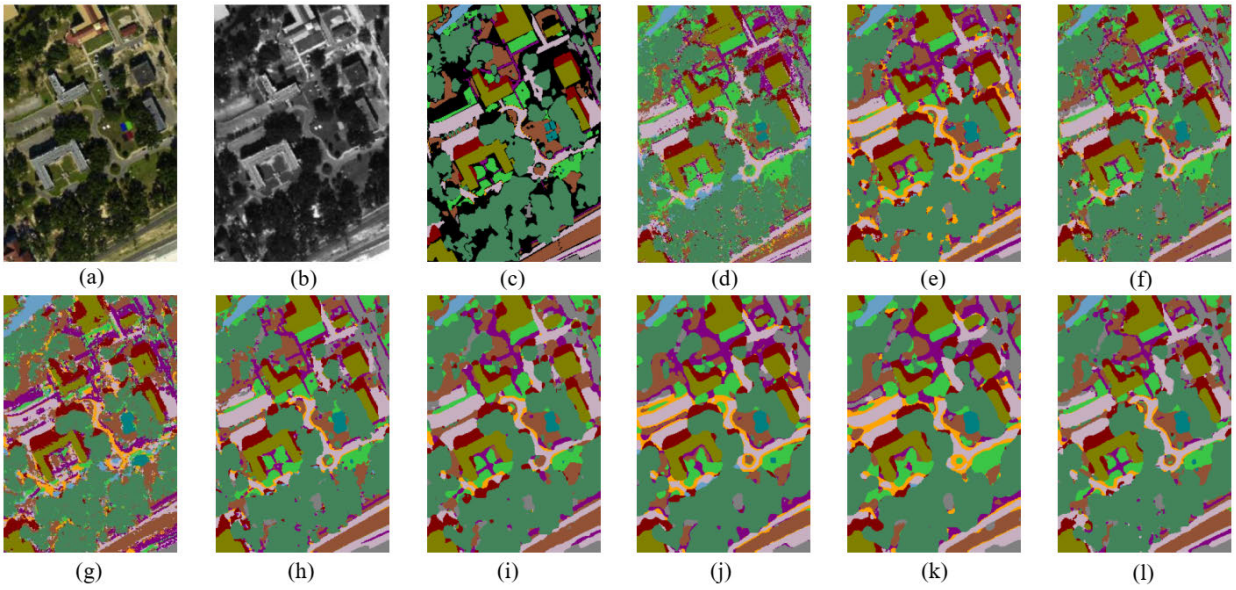


Fig. 7. Classification maps generated by different methods on the MUUFL dataset. (a) Pseudo-color image for HSIs. (b) LiDAR-based DSM. (c) Ground-truth map. (d) ENdNet (87.88%). (e) MDL-RS (89.67%). (f) CCR (90.03%). (g) CALC (90.89%). (h) GLT (92.20%). (i) HCT (91.56%). (j) MFT (86.03%). (k) ExViT (84.47%). (l) Our proposed DSymFuser (93.32%).

TABLE IX
ABLATION ANALYSIS OF DIFFERENT MODAL INPUTS

Input	Houston2013			Trento			MUUFL		
	OA(%)	AA(%)	Kappa ×100	OA(%)	AA(%)	Kappa ×100	OA(%)	AA(%)	Kappa ×100
Only LiDAR	65.34	65.62	62.45	96.86	91.52	95.81	83.36	85.44	78.48
Only HSI	91.96	93.13	91.27	98.02	96.23	97.35	92.25	93.27	89.76
HSI+LiDAR	**96.03**	**96.2**	**95.69**	**99.24**	**98.34**	**98.98**	**93.32**	**94.01**	**91.17**

performance, we conducted three sets of experiments, with single HSI, single LiDAR, and joint HSI and LiDAR as input, respectively. According to the experimental results in Table IX, we find that the multimodal input considerably outperforms the single-modal input in terms of OA, AA, and Kappa metrics on all three datasets. This suggests that there is rich complementary information between different modalities, and multimodal fusion can produce more complete feature representations, bringing gains in classification performance.

2) Effectiveness of Conv3D: To fully exploit the inherent spectral sequential information in HSI data, we utilize Conv3D for spectral feature extraction. To demonstrate the effectiveness

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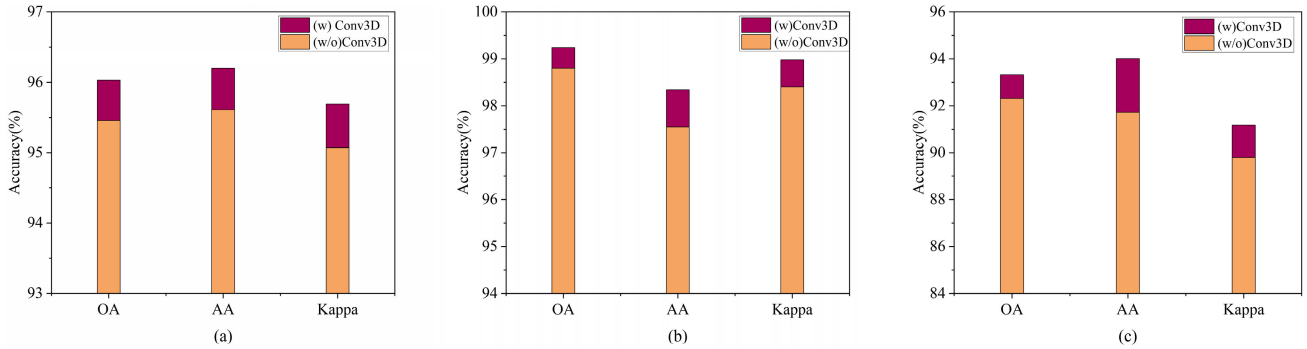


Fig. 8. Effect of Conv3D on classification performance. (a) Houston2013 dataset. (b) Trento dataset. (c) MUUFL dataset.

TABLE X
ABLATION EXPERIMENTS OF LGM ON THE HOUSTON2013 DATASET

HSI		LiDAR		OA	AA	Kappa
Local	Global	Local	Global			
✓	✓	✓		95.76	95.93	95.40
✓	✓		✓	95.64	95.73	95.32
✓		✓	✓	95.67	95.82	95.30
	✓	✓	✓	95.47	95.60	95.08
✓		✓		95.65	95.64	95.28
	✓		✓	94.58	95.00	94.12
✓	✓	✓	✓	**96.03**	**96.20**	**95.69**

of Conv3D, we conducted comparative experiments with and without the Conv3D module, where the experimental results are shown in Fig. 8. We discover from the figure that the inclusion of Conv3D improves the classification performance across all three datasets. Particularly, a more significant performance enhancement is observed on the Houston2013 dataset. It is well known that the Houston2013 dataset exhibits a higher spectral dimensionality compared to the other two datasets, indicating richer spectral information. The result underscores the importance of capturing the spectral information embedded within HSI data, thereby positively impacting the classification performance.

3) Effectiveness of LGM: To break through the limitations of transformers in local feature modeling, we propose LGM to further capture and integrate local–global features from multimodal data. To validate the effectiveness of this module, we conducted two groups of ablation experiments, where the results are shown in Table X. Specifically, the contributions of local features and global features to each modal branch are obtained by replacing LGM with separable convolution and standard transformer, respectively. In addition, the performance with only local features or global features of two modalities is attained as well. From the table, we have below observations: 1) removing global or local features from each modal branch leads to a decrease in classification performance, which justifies the effectiveness of both global and local features in each modality; 2) retaining only global features or local features results in performance degradation as well, this indicates that LGM can effectively capture and integrate local–global features; and 3) the classification performance with only local features is better than with only global features, which demonstrates that local features play a more important role in the MMRSC task.

4) Effectiveness of SFT: To validate the effectiveness of our proposed SFT, we further conducted ablation experiments on different feature interaction patterns. Specifically, we compared SFT with five fusion modes: CNN-based concatenation fusion, self-attention concatenation fusion, self-attention addition fusion, cross-attention fusion, and multiple cross-attention fusion, which are represented by Conv + Cat, SA + Cat, SA + Add, CA, and multiple CA [32], respectively. The experimental results are reported in Table XI. It can be found from the table that the transformers-based methods outperform the CNN-based methods in terms of feature interaction. In addition, compared to existing fusion approaches based on self-attention or cross-attention mechanisms, our proposed SFT exhibits superior fusion performance. This suggests that the symmetric fusion strategy is more adept at integrating complementary information from different modalities. Compared with the cross-attention mechanism, SFT achieves an increase of 0.34% in OA, highlighting the benefits of impartially treating diverse modalities and allowing the network to adaptively learn the optimal fusion strategy.

D. Parameter Analysis

In this section, we analyze the effect of two hyperparameters, that is, depths of model L and patch size p, on the classification performance of the proposed DSymFuser.

1) Depths of Model: We regard the combination of one LGM layer and one SFT layer as a structural unit of the proposed model. To analyze the impact of the depths of the model (layers of structural units) on the classification performance, we conducted experiments with {1, 2, 3, 4, 5} as the model depth candidate set, where the results are illustrated in Fig. 9. According to the experimental results, the optimal model depth for the Houston2013 and MUUFL datasets is 2, and it is 1 for the Trento dataset. With the increase of the model depth, the classification performance deteriorates on all three datasets, which can be explained by the appearance of overfitting in the large model depth scenario.

2) Patch Size: To analyze the impact of the image patch size on the classification performance, we tested on seven

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TABLE XI
PERFORMANCE COMPARISON OF DIFFERENT FUSION MODES ON THE HOUSTON2013 DATASET

Fusion Mode	Conv+Cat	SA+Cat	SA+Add	CA	multiple CA	Ours
OA(%)	95.20	95.30	95.83	95.69	95.13	**96.03**
AA(%)	95.55	95.74	96.03	95.94	95.45	**96.20**
Kappa×100	94.78	94.90	95.50	95.32	94.71	**95.69**

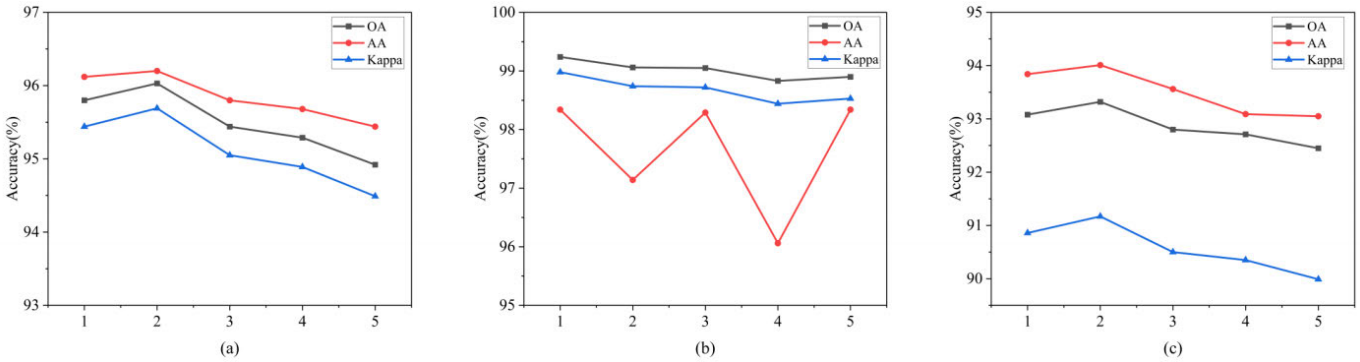


Fig. 9. Influence of depth of model on classification performance. (a) Houston2013 dataset. (b) Trento dataset. (c) MUUFL dataset.

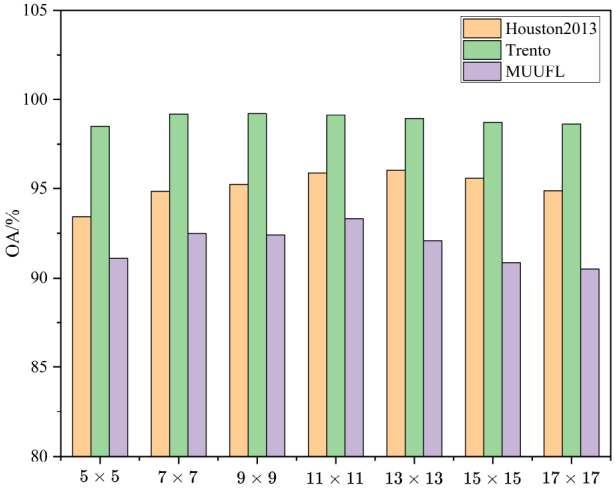


Fig. 10. Effect of different patch sizes on classification performance.

input patch sizes: 5 × 5, 7 × 7, 9 × 9, 11 × 11, 13 × 13, 15 × 15, and 17 × 17. As exhibited in Fig. 10, for the Houston2013, Trento, and MUUFL datasets, the highest accuracy values are achieved with patch sizes 13 × 13, 9 × 9, and 11 × 11, respectively. This reveals that medium-sized patches are more preferential for representation learning, for they achieve an appropriate compromise between contextual information involvement and feature redundancy.

V. CONCLUSION AND FUTURE WORK

In this article, we put forward a dual-branch transformer-based framework, called DSymFuser for the MMRSC task. Aiming at extracting discriminative single-modal features, we designed an LGM module to take full advantage of the local modeling ability of CNNs and the long-range modeling capability of transformers. To adequately exploit the interactions between modalities, we developed an SFT, where the symmetric query manner excavates more comprehensive cross-modal relationships. By sequentially stacking the combinations of LGM and SFT modules, multilevel local–global features and cross-modal relationships can be simultaneously obtained. Extensive ablation studies and comparative experiments demonstrated the effectiveness and superiority of DSymFuser. It should also be highlighted that the proposed LGM and SFT modules are both portable blocks that can be flexibly applied to other multimodal data processing tasks.

Label scarcity has been a longstanding issue confronted with the MMRSC task. Therefore, we plan to develop self-supervised or semi-supervised multimodal data classification frameworks in the future, to fully leverage the massive unlabeled data. In addition, we are interested in designing networks with adaptive local receptive fields, to address the class confusion issue between spatially adjacent classes.

REFERENCES

[1] R. Fei et al., "Retrieval of the aerosol extinction coefficient from scanning scheimpflug LiDAR measurements for atmospheric pollution monitoring," *Atmos. Environ.*, vol. 309, Sep. 2023, Art. no. 119945.
[2] A. M. Coutts, R. J. Harris, T. Phan, S. J. Livesley, N. S. G. Williams, and N. J. Tapper, "Thermal infrared remote sensing of urban heat: Hotspots, vegetation, and an assessment of techniques for use in urban planning," *Remote Sens. Environ.*, vol. 186, pp. 637–651, Dec. 2016.
[3] J. A. Fortin, J. A. Cardille, and E. Perez, "Multi-sensor detection of forest-cover change across 45 years in Mato Grosso, Brazil," *Remote Sens. Environ.*, vol. 238, Mar. 2020, Art. no. 111266.
[4] S.-H. Lee, K.-J. Han, K. Lee, K.-J. Lee, K.-Y. Oh, and M.-J. Lee, "Classification of landscape affected by deforestation using high-resolution remote sensing data and deep-learning techniques," *Remote Sens.*, vol. 12, no. 20, p. 3372, Oct. 2020.
[5] W. Han et al., "A survey of machine learning and deep learning in remote sensing of geological environment: Challenges, advances, and opportunities," *ISPRS J. Photogramm. Remote Sens.*, vol. 202, pp. 87–113, Aug. 2023.
[6] D. I. Enderle and R. C. Weih Jr., "Integrating supervised and unsupervised classification methods to develop a more accurate land cover classification," *J. Arkansas Acad. Sci.*, vol. 59, no. 1, pp. 65–73, 2005.

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[7] M. Carranza-García, J. García-Gutiérrez, and J. Riquelme, "A framework for evaluating land use and land cover classification using convolutional neural networks," *Remote Sens.*, vol. 11, no. 3, p. 274, Jan. 2019.

[8] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[9] M. Letard et al., "3DMASC: Accessible, explainable 3D point clouds classification. Application to bi-spectral topo-bathymetric LiDAR data," *ISPRS J. Photogramm. Remote Sens.*, vol. 207, pp. 175–197, Jan. 2024.

[10] Z. Kuang, H. Bi, F. Li, C. Xu, and J. Sun, "Polarimetry-inspired contrastive learning for class-imbalanced PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5212819.

[11] C. Weitkamp, *LiDAR: Range-Resolved Optical Remote Sensing of the Atmosphere*, vol. 102. Berlin, Germany: Springer, 2006.

[12] J. Jung, E. Pasolli, S. Prasad, J. C. Tilton, and M. M. Crawford, "A framework for land cover classification using discrete return LiDAR data: Adopting pseudo-waveform and hierarchical segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 2, pp. 491–502, Jan. 2014.

[13] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.

[14] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[15] L. Gómez-Chova, D. Tuia, G. Moser, and G. Camps-Valls, "Multimodal classification of remote sensing images: A review and future directions," *Proc. IEEE*, vol. 103, no. 9, pp. 1560–1584, Sep. 2015.

[16] D. Hong, C. Li, B. Zhang, N. Yokoya, J. A. Benediktsson, and J. Chanussot, "Multimodal artificial intelligence foundation models: Unleashing the power of remote sensing big data in Earth observation," *Innov. Geosci.*, vol. 2, no. 1, 2024, Art. no. 100055.

[17] S. Morchhale, V. P. Pauca, R. J. Plemmons, and T. C. Torgersen, "Classification of pixel-level fused hyperspectral and LiDAR data using deep convolutional neural networks," in *Proc. 8th Workshop Hyperspectral Image Signal Processing, Evol. Remote Sens. (WHISPERS)*, Aug. 2016, pp. 1–5.

[18] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017.

[19] C. Ge, Q. Du, W. Sun, K. Wang, J. Li, and Y. Li, "Deep residual network-based fusion framework for hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2458–2472, 2021.

[20] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017.

[21] J. Deng et al., "RustQNet: Multimodal deep learning for quantitative inversion of wheat stripe rust disease index," *Comput. Electron. Agricult.*, vol. 225, Oct. 2024, Art. no. 109245.

[22] C. Li et al., "CasFormer: Cascaded transformers for fusion-aware computational hyperspectral imaging," *Inf. Fusion*, vol. 108, Aug. 2024, Art. no. 102408.

[23] N. Audebert, B. Le Saux, and S. Lefèvre, "Beyond RGB: Very high resolution urban remote sensing with multimodal deep networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 140, pp. 20–32, Jun. 2018.

[24] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[25] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[26] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[27] Y. Zhang et al., "Multimodal transformer network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514317.

[28] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.

[29] M. Zhou et al., "A general spatial-frequency learning framework for multimodal image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, 2024, doi: 10.1109/TPAMI.2024.3368112.

[30] B. Tu, Q. Ren, J. Li, Z. Cao, Y. Chen, and A. Plaza, "NCGLF2: Network combining global and local features for fusion of multisource remote sensing data," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102192.

[31] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[32] X. Chen, B. Yan, J. Zhu, D. Wang, X. Yang, and H. Lu, "Transformer tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8126–8135.

[33] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[34] W. Liao, A. Pižurica, R. Bellens, S. Gautama, and W. Philips, "Generalized graph-based fusion of hyperspectral and LiDAR data using morphological features," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 552–556, Mar. 2015.

[35] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[36] M. Khodadadzadeh, J. Li, S. Prasad, and A. Plaza, "Fusion of hyperspectral and LiDAR remote sensing data using multiple feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2971–2983, Jun. 2015.

[37] D. Hong, X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. X. Zhu, "Invariant attribute profiles: A spatial-frequency joint feature extractor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3791–3808, Jun. 2020.

[38] C. Ge, Q. Du, W. Li, Y. Li, and W. Sun, "Hyperspectral and LiDAR data classification using kernel collaborative representation based residual fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1963–1973, Jun. 2019.

[39] Y. Gu, Q. Wang, X. Jia, and J. A. Benediktsson, "A novel MKL model of integrating LiDAR data and MSI for urban area classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5312–5326, Oct. 2015.

[40] L. Dabbiru, S. Samiappan, R. A. A. Nobrega, J. A. Aanstoos, N. H. Younan, and R. J. Moorhead, "Fusion of synthetic aperture radar and hyperspectral imagery to detect impacts of oil spill in Gulf of Mexico," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 1901–1904.

[41] B. Rasti, P. Ghamisi, and R. Gloaguen, "Hyperspectral and LiDAR fusion using extinction profiles and total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3997–4007, Jul. 2017.

[42] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2018.

[43] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[44] A. Pournemat, P. Adibi, and J. Chanussot, "Semisupervised charting for spectral multimodal manifold learning and alignment," *Pattern Recognit.*, vol. 111, Mar. 2021, Art. no. 107645.

[45] Y. Gao, M. Zhang, W. Li, X. Song, X. Jiang, and Y. Ma, "Adversarial complementary learning for multisource remote sensing classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505613.

[46] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[47] H.-C. Li, W.-S. Hu, W. Li, J. Li, Q. Du, and A. Plaza, "A3 CLNN: Spatial, spectral and multiscale attention ConvLSTM neural network for multisource remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 747–761, Feb. 2022.

[48] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2021.

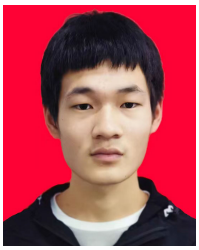
[49] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.

[50] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

[51] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.

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[52] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[53] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1251–1258.
[54] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, Aug. 2020.
[55] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[56] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.
[57] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2020.
[58] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.



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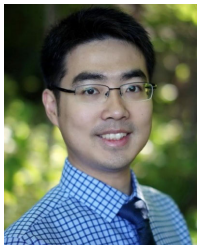
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